# COMP3423 Human Computer Interaction Assignment 1

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## 1 User Research

### 1.1 Analysis of an Existing System

In this section, I will analyze the user interface of a mobile phone self-service ordering system of the food court of Fortune Metropolis (link). The food court is located within a large commercial complex in Hung Hom, consisting of seven distinct eateries sharing a communal dining area, where all orders to the eateries are placed through a self-service ordering system. Paragraphs below will discuss the system's shortcomings regarding the user interface design based on my personal experience.

**Unreliable Device Adaptablity** The system is designed to be used on mobile phones by scanning a QR code. However, since the system is based on a web page that is not specifically optimized for mobile devices with distinct screen sizes, the user interface may not be displayed properly on web browser. As shown in Figure 1 and Figure 2, the welcome page is not fully displayed on the screen without automatic centering, and users need to scroll vertically or horizontally to view the entire page. Moreover, in comparison to the screen size, the interactive buttons are small that can lead to the fat finger problem.

More significantly, due to the designer's oversight of the native UI of mobile browsers, some crucial buttons get obscured, resulting in customer confusion. For instance, the back button that can be seen in Figure 3 is blocked by the Safari browser's navigation bar as shown in Figure 4. These factors may cause customers to perceive the pages as unattractive and not user-friendly.

**Eatery Menu Seperation** If customers can not find the desired food in the first eatery or would like to order across eateries, they are required to return to the eatery selection page in Figure 4 and reselect. In contrast, the ordering system on the self-service kiosk in the food court allows customers to browse the menus of all eateries on the same page. The separation of the menu browsing and eatery selection pages force customers to switch between pages frequently.

**Mixed Interactive Logic** After selecting an eatery, the menu integrates both the vertical and horizontal interactive logic, that is, customers are required to scroll horizontally to view the categories of food and vertically to view the food items in each category as shown in Figure 5. Since most customers are accustomed to the vertical scrolling logic on mobile devices, and the system does not provide a clear indication of the horizontal scrolling, customer may neglect the horizontal scrolling and miss some food categories or find it annoying to scroll in two directions.

**Complicated Option Lists** Lists of options for a food item are supposed to be chosen by customers before adding the item to the cart. In Figure 6, the options cover the entire screen, but the indications for the categorization of options (the different lists) are not prominent, which may lead to the following issues:

- Too much information, causing customers to spend a longer time on a single interface for processing the options.
- Customers may select multiple options within the same category due to unclear categorization, but that category may require single selection, resulting in order changes.
- Mandatory and non-mandatory lists are intertwined, causing frustration for customers when adding items to the cart as the system prompts that mandatory options are incomplete, leading to the customers' impatience.

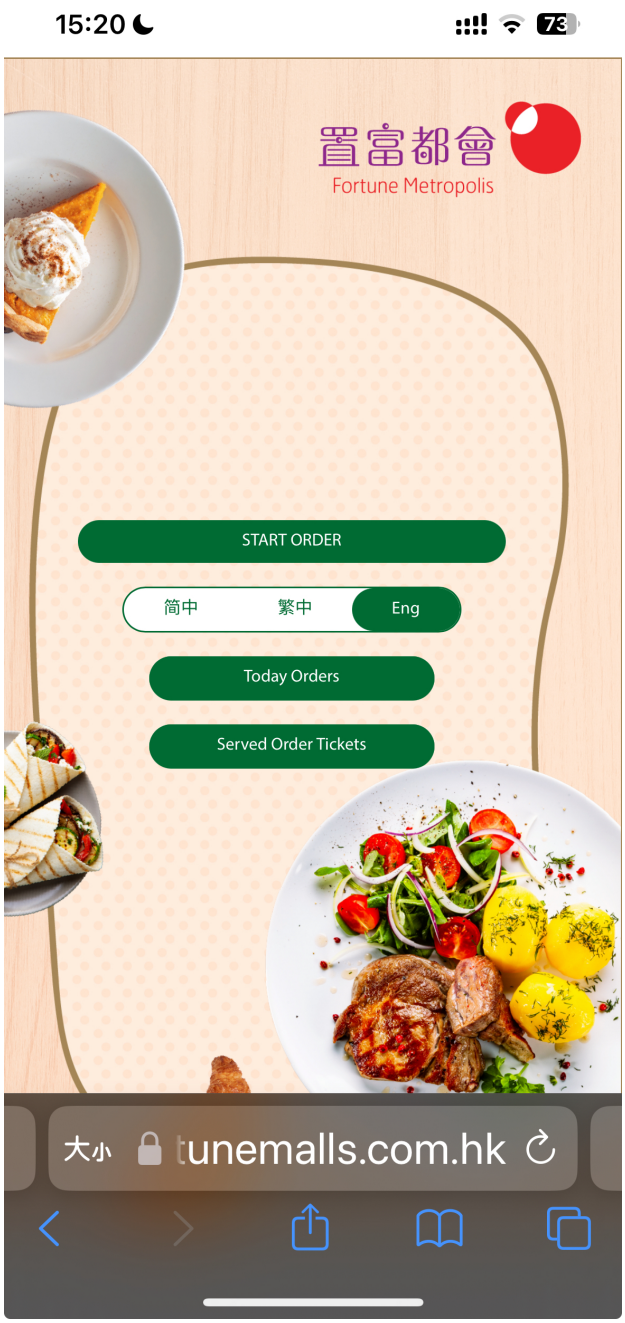



Figure 1: The welcome page scrolled left, not fully displayed & centered




Figure 2: The welcome page scrolled right with the same problem

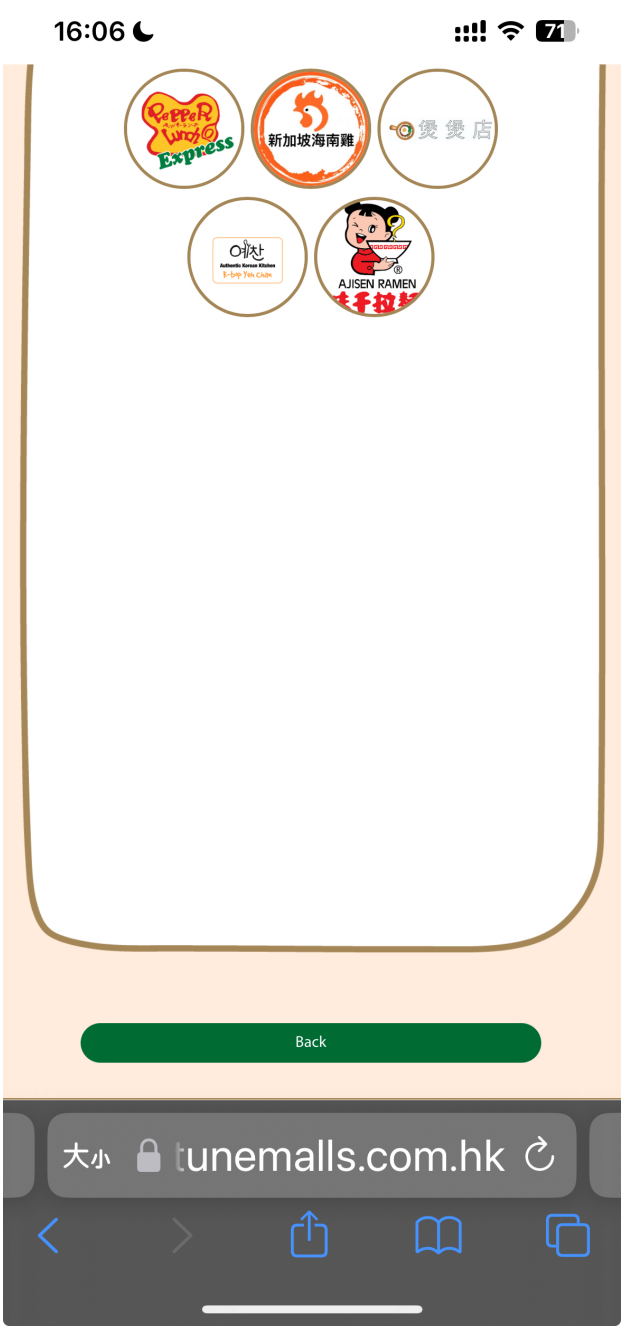



Figure 3: The back button can be seen

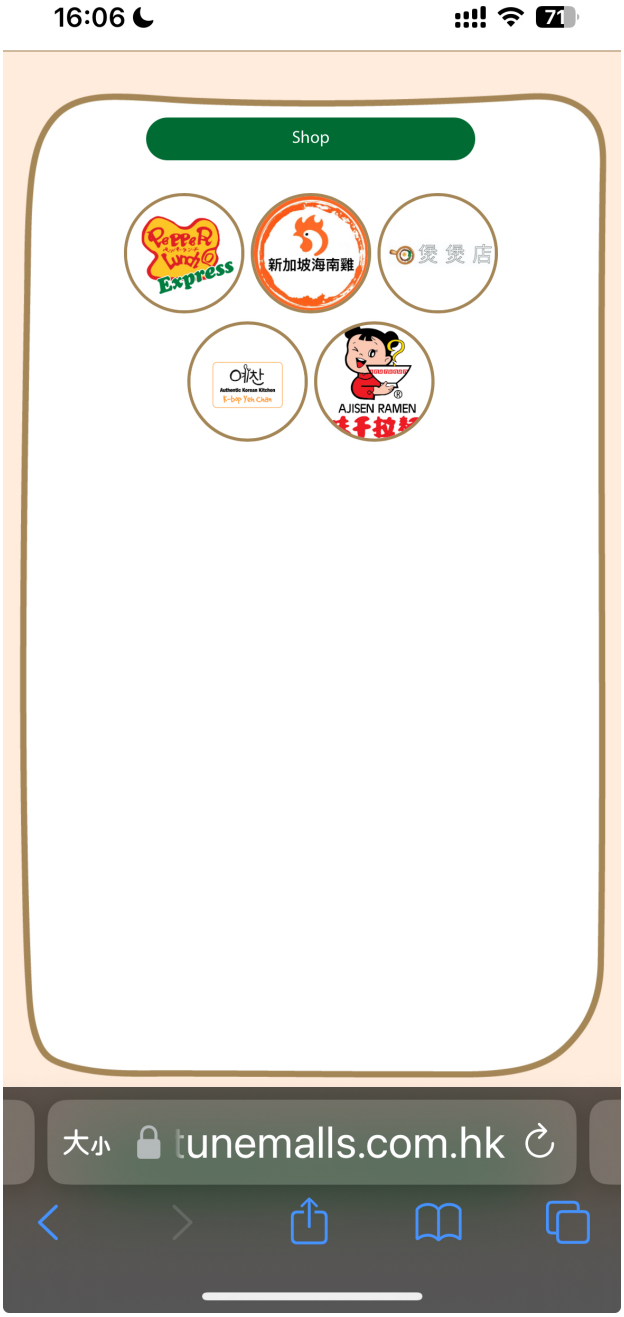



Figure 4: The back button is blocked by the navigation bar

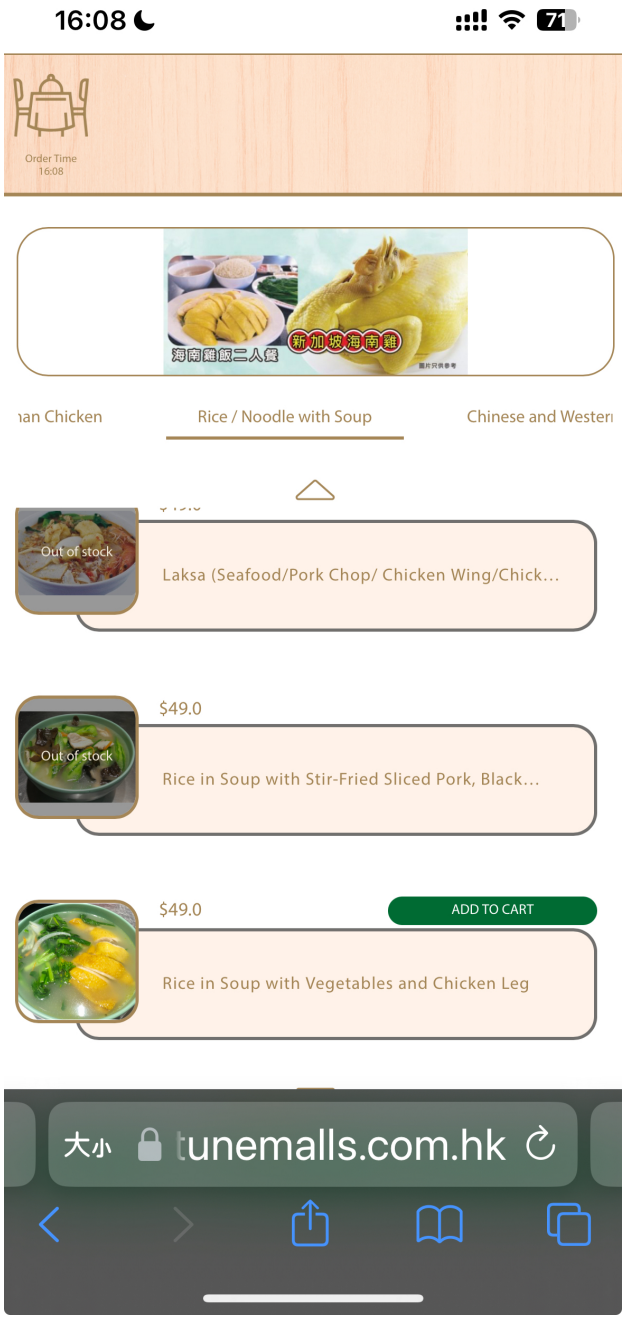



Figure 5: The mixed interactive logic in the menu page




Figure 6: The Complicated option lists

**Style Inconsistency** The page for served order tickets in Figure 7 adapts purple as the primary color, differing from the previous pages that use green as the primary color. The fonts are also different.




Figure 7: The inconsistent style in the served order tickets page

## 1.2 User Interview

One system that I would like to discuss is the self-service ordering system of the food court of Fortune Metropolis. Her responses are as follows":

- "When checking out my shopping cart, I sometimes forget whether I selected "dine-in" or "takeaway" in the previous step, and the system does not provide a prompt. To confirm, I have to abandon all the items already selected and exit to the welcome page to reorder. I may also forget to ask myself this question and only realize the mistake when collecting my food."
- "Additionally, there are times when I want to order one item for dining in at the food court and another to take away for a friend, but it is not feasible within the same order for this system. I have to place

multiple orders."

- "The text is small and too densely packed. I hate zooming in and out from time to out."
- "There are few and small-sized pictures of the food items, and since I haven't tried some of the dishes, I can only rely on the images to make my decision."
- I can not search a specific food item in the system, and I have to browse through the entire menu to find it.

Therefore, five user requirements can be derived from the interview to enhance the usability of the current ordering system:

- The system should provide a user with an indication of the dining method selected in the shopping cart, and a method to change the dining method after the first selection.
- The system should allow a user to choose the dining method for each food item in the same order.
- The system should provide a readable text design for users that is at least 12pt in font size.
- The system should display multiple and large pictures of the food items.
- The system should be implemented with a search function for users to find specific food items by typing relevant information.

### 1.3 User Persona

**Name** Billy

**Motivation** He often visits the food court of Fortune Metropolis for lunch and dinner with his friends to enjoy the variety of food options since it is near to his university.

**Goals** To find and order his desired food items quickly and accurately using the self-service ordering system on his mobile phone.

**Pains** Suffers from the inconvenience of the current system not aligned with user intuition in interactive design and lack of functions of searching food items and editing order details.

## 2 Wireframing

### 2.1 Usability Goals

**Redesign Objectives** Three important redesign objectives are listed below:

- To improve the adaptability on mobile devices by optimizing the layout, button size, typographical and interactive design.
- To integrate both the menu browsing and category selection pages and the shopping cart page and functions of changing dining method to reduce the frequency of page switching with intuitive interactive logic.
- To provide a clear and attractive detail interface for each food item that integrates several large high-resolution images and text information to help users learn more about the food items and make decisions.

**Prioritized Requirements** Five prioritized requirements are listed below:

- The system should provide usable pages and interactions for the integration of functions including browsing menus, obtaining information of each food item, adding food items to a shopping cart, and making an order.
- The static layouts should be adjusted to adapt to mobile devices for the sake of stability.
- The navigation flow and the dynamic interactive logic should be intuitive to reduce the cognitive load of users.
- Additional functions such as searching and shopping cart and dining method management should be seamlessly embedded along with appropriate interactive designs.
- TThe overall design should be aesthetically pleasing and consistent in style to enhance the visual appeal of the system.

### 2.2 Low-Fidelity Wireframes

The low-fidelity wireframes are appended at the end.

### 2.3 Navigation Flow

*Welcome Page → Start → The first menu page → Select Food Category → The second menu page → Choose food item → Specific food item page → Add to cart → Shopping cart → Adjust quantity → Choose the dining method → Payment*

In this flow, if the desired food item is already on the first menu page, the user can skip the second menu page and directly select the food item. If a user is familiar with a food item, the specific food item page can also be skipped. Meanwhile, the function of searching can be reached from menu pages, and the function of managing shopping cart can be reached from the shopping cart page.

## 3 Visual Design and Prototyping

### 3.1 Style Guide

**Color Scheme** #D68337 with 70% opacity is chosen as the primary color in the visual design of the system, which is a warm and vibrant color that can attract users' attention and stimulate their appetite. It is mainly used for buttons, icons and frame strokes. #28C71A with 30% opacity is chosen as the color for special buttons and icons such as the shopping cart and payment button. The background color is white to ensure the readability of the text and the clarity of the images.

**Typography** *Kodchasan* is chosen as the primary font for the system, which is a modern and clean sans-serif font that is easy to read on screens. Regular weight is used for information text, medium weight is used for subheadings and buttons, and semibold weight is used for headings. The font size is set to 24 pt for headings, subheadings, and buttons and 16 pt for information text.

**Icongraphy** Only icons drawn in black with scalable vector graphics (.svg) are used in the system to ensure the consistency of the visual style.

**Imagery** High-resolution images of food items are used in the system to provide users with a clear and realistic view of the dishes. The images are displayed in rectangle frames with a border to maintain consistency. A large image with layer blur effect and 50% opacity is used as the theme image covering the welcome page and also as the half-background image for all other pages.

### 3.2 Prototyping

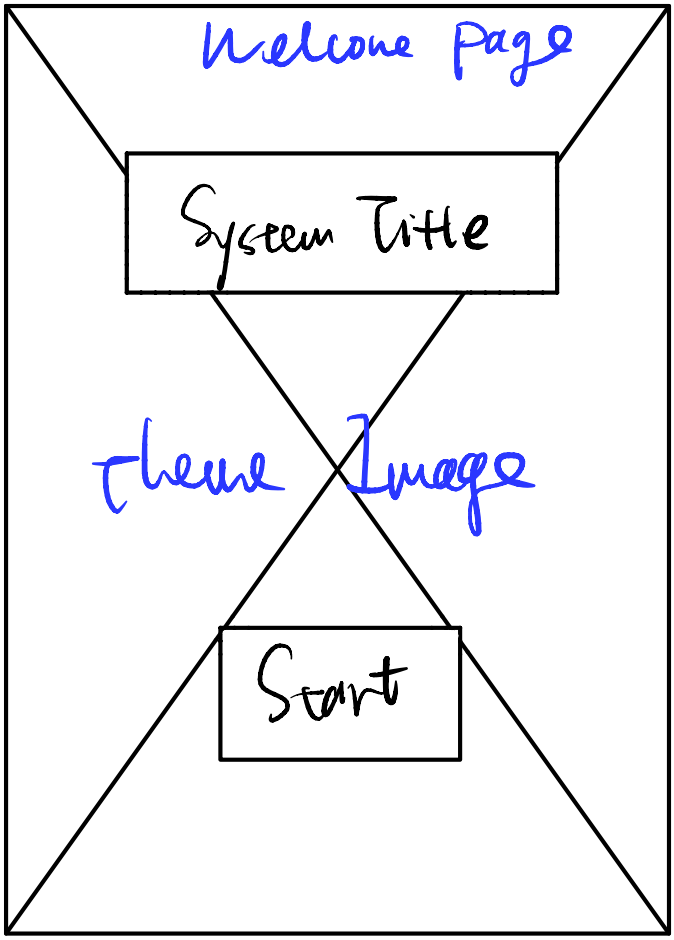
The Figma prototype is available at this link. Hereby, screen captures of the prototype are listed at the end.

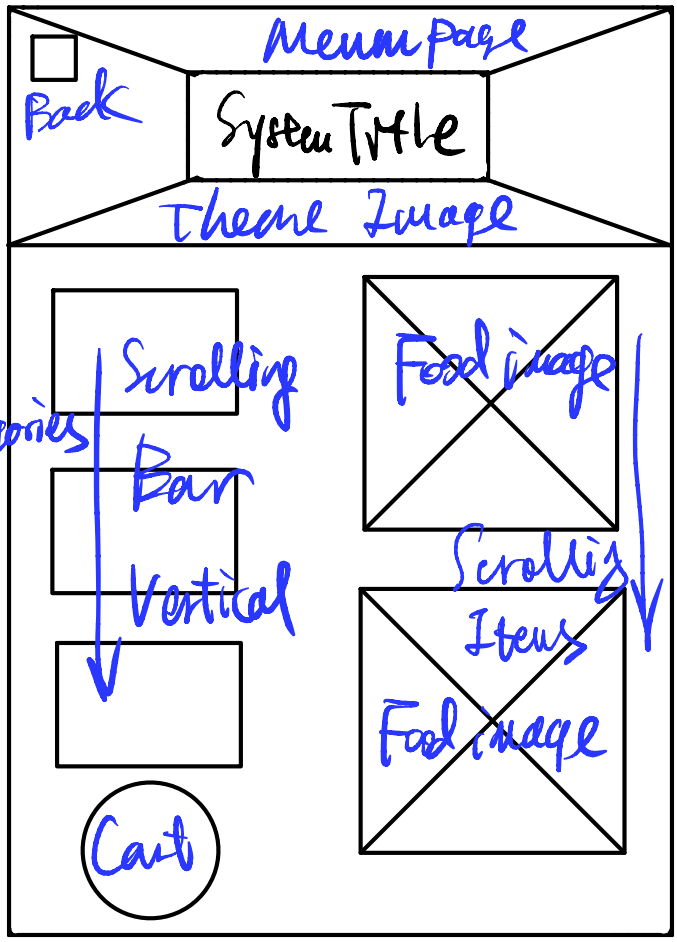
## 4 Usability Testing

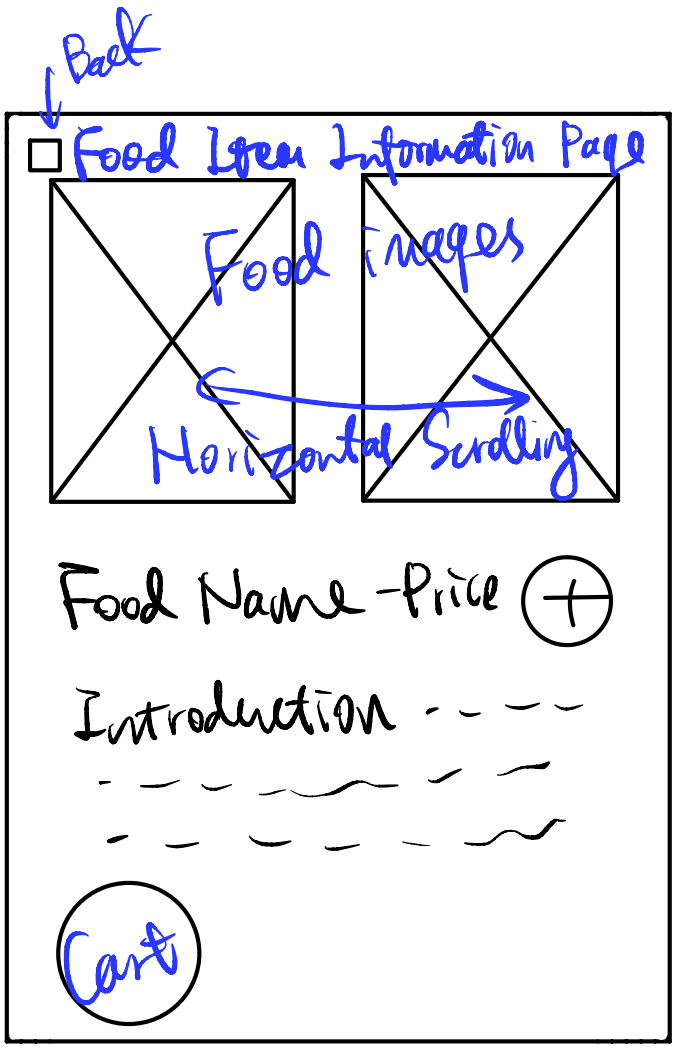
Seven individual reviewers were invited to participate in the usability testing on *maze.co* of the prototype. The task was to try browsing as many pages as possible and eventually enter the shopping cart page. In that case, most of interactive designs were validated, and two findings were discovered by feedbacks from the reviewers:

**Welcome Page Confusion** The *Start* button on the welcome page was not prominent enough, as three reviewers tried to click the title frame before they clicked the button. To ensure the consistency of the color scheme, the button adapts the same color as the title frame, which may cause confusion for users that the title frame is also clickable. As a result, users might spend more time on the welcome page. To improve the interaction efficiency the title frame is supposed to be enlarged and the effects for differentiate the button from the frame should be added, such as shadow effects.

**Unintuitive Interavtive Design** In the shopping cart page, a vertical scrolling logic is attached to each food item for users to select the dining method. However, five reviewers spent more that 15 seconds to understand the scrolling logic and then select the dining method, as the logic is not widely adapted currently in the mobile applications. The original goal of the scrolling logic is to avoid the fat finger problem, but it may cause the extension of the interaction time, especially for new users. This can be regarded as a trade-off between accuracy and efficiency.


Welcome Page
System Title
theme Image
Start


Menu Page
Back
System Title
Theme Image
Scrolling
Categories
Bar
Vertical
Food image
Scrolling
Items
Food image
Cart


Back
Food Item Information Page
Food Images
Horizontal Scrolling
Food Name - Price
Introduction
Cart

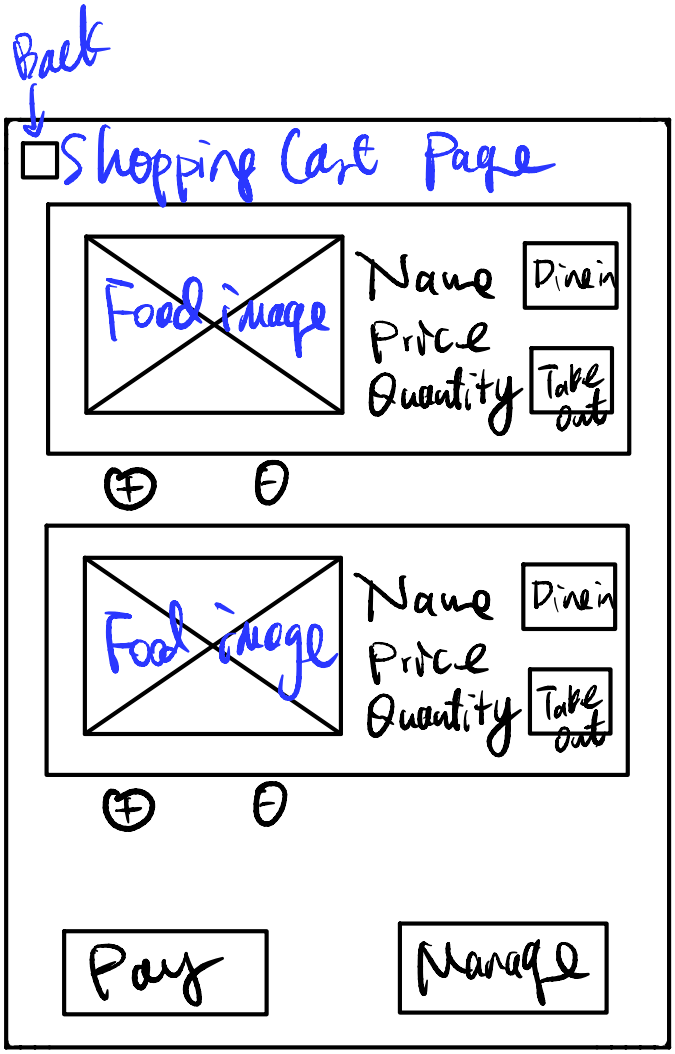

Back
Shopping Cart Page
Food image
Name
Price
Quantity
Dine in
Take out
Food image
Name
Price
Quantity
Dine in
Take out
Pay
Manage




Figure 8: The welcome page




Figure 9: One of the menu pages




Figure 10: The menu page after frame scrolling

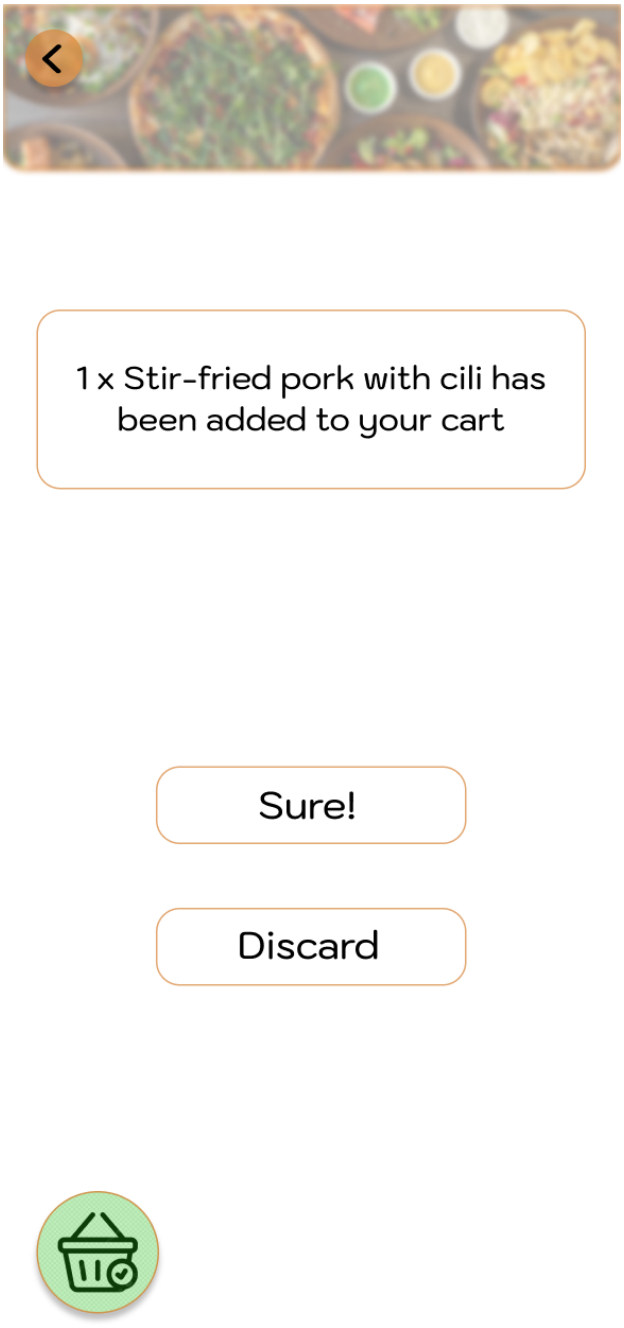



Figure 11: The confirmation page for the selected item

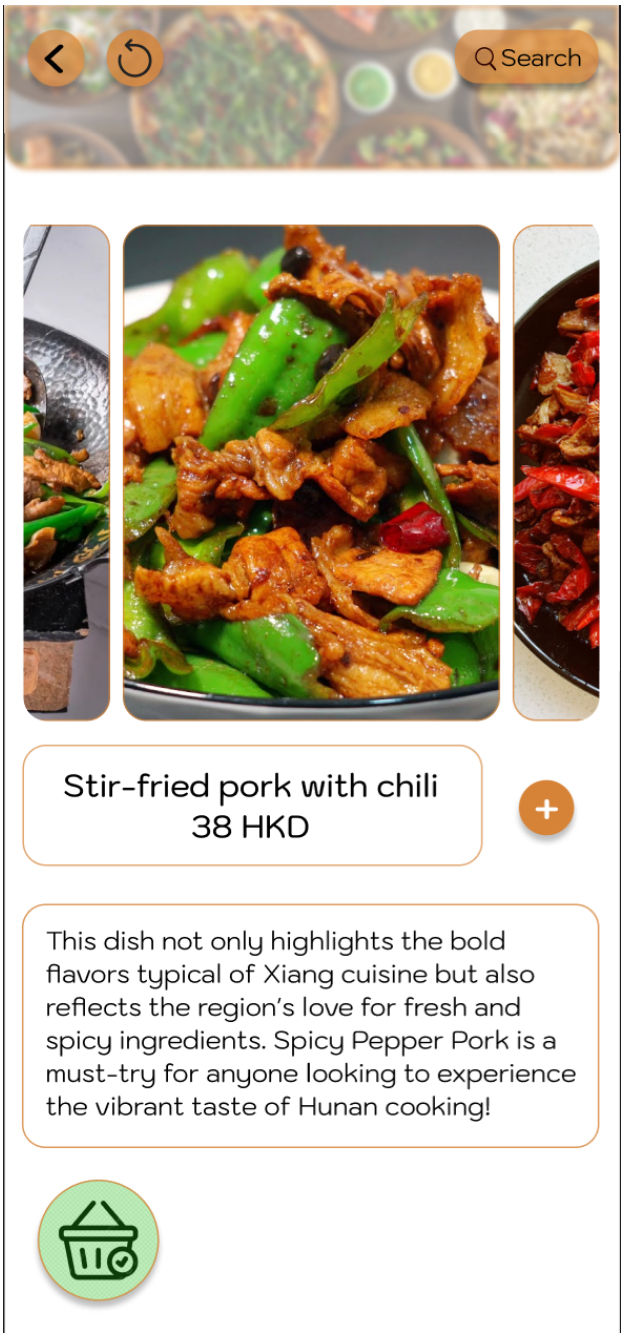



Figure 12: The page for a specific food item

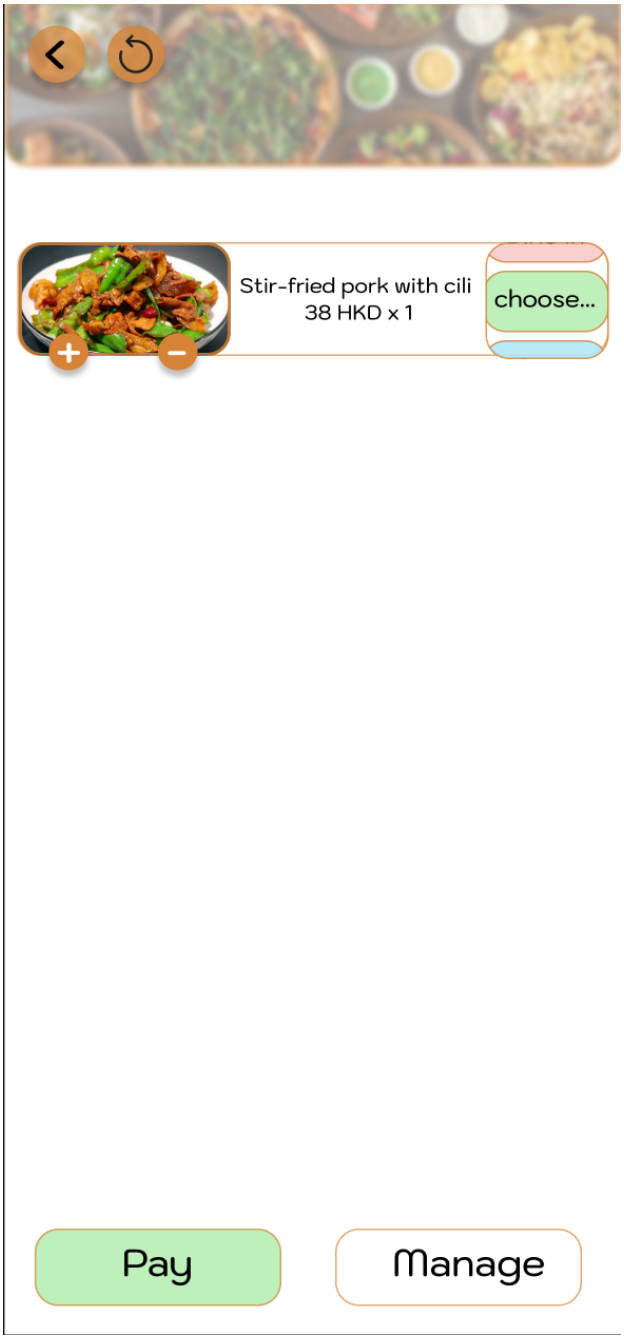



Figure 13: The shopping cart page